

Duality II: Optimality and Complementary Slackness

A feasible solution and a feasible dual bound can meet. When they do, we have a complete optimality certificate.

Our route: strong duality $\rightarrow$ complementary slackness $\rightarrow$ the simplex connection.

We continue with minimization problems; their duals maximize lower bounds.

Practice writing duals

Strong duality

Complementary slackness

Recover the dual variables

Optional proofs and extensions

How to construct the dual

- 1. Objective:** a minimization primal $\min c^\top x$ gives a maximization dual $\max b^\top p$.
- 2. One dual variable per row.** Row i of A is $a_i^\top$; its constraint sets the sign of p_i .

Primal constraint	Dual-variable restriction
$a_i^\top x \geq b_i$	$p_i \geq 0$
$a_i^\top x \leq b_i$	$p_i \leq 0$
$a_i^\top x = b_i$	p_i unrestricted

- 3. One dual constraint per variable.** $(A^\top p)_j$ is the dual left-hand side for x_j .

Primal-variable restriction	Dual constraint
$x_j \geq 0$	$(A^\top p)_j \leq c_j$
$x_j \leq 0$	$(A^\top p)_j \geq c_j$
x_j unrestricted	$(A^\top p)_j = c_j$

- 4. Check:** include the objective, every constraint, and every variable restriction.

Practice: write the complete dual

8–10 minutes. Work individually or with a partner. Include the objective, constraints, and variable restrictions. You do not need to solve the problems.

Problem A

$$\begin{array}{ll}\min & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4, \quad x_1 + 2x_2 \geq 6, \quad x_1, x_2 \geq 0.\end{array}$$

Problem B

$$\begin{array}{ll}\min & x_1 + 2x_2 \\ \text{s.t.} & x_1 - x_2 \leq 3, \quad 2x_1 + x_2 \geq 2, \\ & x_1 \text{ unrestricted}, \quad x_2 \geq 0.\end{array}$$

Problem C

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \quad 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

All three primals minimize. Use the sign guide on the previous page; answers follow.

Problem A: nonnegative variables and $\geq$ rows

Problem A

$$\begin{array}{ll}\min & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4, \quad x_1 + 2x_2 \geq 6, \quad x_1, x_2 \geq 0.\end{array}$$

Dual

$$\begin{array}{ll}\max & 4p_1 + 6p_2 \\ \text{s.t.} & p_1 + p_2 \leq 2, \\ & p_1 + 2p_2 \leq 3, \quad p_1, p_2 \geq 0.\end{array}$$

Dual-variable signs: both primal rows are $\geq$, so $p_1, p_2 \geq 0$.

Dual constraints: both primal variables are nonnegative, so both dual left-hand sides must be $\leq$ their cost coefficients.

Problem B: mixed rows and a free variable

Problem B

$$\begin{array}{ll}\min & x_1 + 2x_2 \\ \text{s.t.} & x_1 - x_2 \leq 3, \quad 2x_1 + x_2 \geq 2, \\ & x_1 \text{ unrestricted}, \quad x_2 \geq 0.\end{array}$$

Dual

$$\begin{array}{ll}\max & 3p_1 + 2p_2 \\ \text{s.t.} & p_1 + 2p_2 = 1, \\ & -p_1 + p_2 \leq 2, \quad p_1 \leq 0, \quad p_2 \geq 0.\end{array}$$

Dual-variable signs: row 1 is $\leq$, so $p_1 \leq 0$; row 2 is $\geq$, so $p_2 \geq 0$.

Dual constraints: unrestricted x_1 requires equality; $x_2 \geq 0$ gives a $\leq$ comparison.

Problem C: equality rows and a matching bound

Problem C

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \quad 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

Dual

$$\begin{array}{ll}\max & 8p_1 + 3p_2 \\ \text{s.t.} & 5p_1 + 3p_2 \leq 13, \quad p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \text{ unrestricted.}\end{array}$$

Dual-variable signs: equality rows allow unrestricted dual variables. **Dual constraints:** all $x_j \geq 0$, so all three comparisons are $\leq$.

The feasible points $x = (1, 0, 1)$ and $p = (2, 1)$ both have value 19. Weak duality certifies optimality; the appendix contains the detailed checks.

Next question: if a primal has a finite optimum, must a matching dual solution always exist?

Strong duality: the best bound reaches the best cost

$$\begin{aligned} (P) \quad & \min c^\top x && \text{s.t. } Ax = b, \ x \geq 0, \\ (D) \quad & \max b^\top p && \text{s.t. } A^\top p \leq c, \ p \text{ unrestricted.} \end{aligned}$$

Theorem 4.4 — assumption: the primal LP is feasible and has a finite optimal value.

Conclusion: both primal and dual attain optimal solutions, and their optimal objective values are equal.

$$c^\top x^* = b^\top p^*.$$

The same statement holds starting from a finite dual optimum. Strong duality does not require nondegeneracy.

What can happen to a primal–dual pair?

Primal minimization	Dual finite optimum	Dual unbounded above	Dual infeasible
Finite optimum	Possible: equal values	Impossible	Impossible
Unbounded below	Impossible	Impossible	Possible
Infeasible	Impossible	Possible	Possible

A finite optimum on one side gives a finite optimum on the other. If one problem is unbounded, the other must be infeasible.

An infeasible primal does not by itself prove that the dual is unbounded. [See an example in which both are infeasible.](#)

The final primal basis gives an optimal dual solution

Primal: minimize $c^\top x$ subject to $Ax = b, x \geq 0$.

Dual: maximize $b^\top p$ subject to $A^\top p \leq c, p$ unrestricted.

p_i is the **dual variable for primal equality** i ; it can have either sign.

At simplex termination, let x^* be the primal basic solution. B contains its basic columns; $c_{\mathcal{B}}$ contains their objective coefficients.

Find the dual solution: solve

$$B^\top p^* = c_{\mathcal{B}}.$$

$$A^\top p^* \leq c, \quad b^\top p^* = c^\top x^*.$$

These dual-variable values satisfy every dual constraint and give the same objective value as the primal solution.

Both solutions are optimal. See the appendix for verification.

Complementary slackness: when are two feasible solutions optimal?

Strong duality: when a finite optimum exists, an optimal primal–dual pair with equal values exists.

Our new question: given any primal feasible x and dual feasible p , what conditions tell us that both are optimal?

The goal is to recognize optimality directly from variables and constraint slacks.

Complementary slackness: for a feasible pair to be optimal, each variable and its corresponding dual slack must have product zero. Inequality constraints add a second pairing.

A dual slack is the right-hand side minus the left-hand side of a dual inequality. First see the pairing in our earlier example.

Compute the slack in each dual constraint

Dual: maximize $8p_1 + 3p_2$ subject to
 $5p_1 + 3p_2 \leq 13, p_1 + p_2 \leq 10, 3p_1 \leq 6; p_1, p_2$ unrestricted.

Dual slack = right-hand side – left-hand side. It is nonnegative at a dual feasible solution.

At the dual feasible solution $p = (2, 1)$, compute each slack r_j :

Dual constraint	Left-hand side	Dual slack
$5p_1 + 3p_2 \leq 13$	$5(2) + 3(1) = 13$	$r_1 = 13 - 13 = 0$
$p_1 + p_2 \leq 10$	$2 + 1 = 3$	$r_2 = 10 - 3 = 7$
$3p_1 \leq 6$	$3(2) = 6$	$r_3 = 6 - 6 = 0$

Active means satisfied as an equality. Constraints 1 and 3 are active. Constraint 2 has positive slack.

Next, pair each dual slack with its corresponding primal variable.

Equality constraints: pair each variable with its dual slack

Primal: minimize $c^\top x$ subject to $Ax = b, x \geq 0$.

Dual: maximize $b^\top p$ subject to $A^\top p \leq c, p$ unrestricted.

For a **feasible pair**, define dual slack $r_j = c_j - A_j^\top p \geq 0$. Here A_j is column j .

$$\begin{aligned} c^\top x - b^\top p &= c^\top x - p^\top Ax \\ &= (c - A^\top p)^\top x = \sum_j r_j x_j. \end{aligned}$$

Every term is nonnegative. The gap is zero exactly when $x_j r_j = 0$ for every j .

Therefore a feasible pair is optimal exactly when these products vanish. If $x_j > 0$, its dual inequality must be active; if $r_j > 0$, then $x_j = 0$.

Inequality constraints introduce another pairing

Now change the form: primal constraints may have slack. The dual variables are now nonnegative.

Primal: minimize $c^\top x$ subject to $Ax \geq b, x \geq 0$.

Dual: maximize $b^\top p$ subject to $A^\top p \leq c, p \geq 0$.

The **primal slack** is $s_i = a_i^\top x - b_i \geq 0$; pair it with the dual variable p_i .

The **dual slack** is $r_j = c_j - A_j^\top p \geq 0$; pair it with the primal variable x_j .

With equality constraints, $s_i = 0$ automatically and p_i is unrestricted. With $\geq$ constraints, both s_i and p_i are nonnegative.

The next page explains why both kinds of products appear in the objective gap.

Why both sets of complementary products must vanish

For the inequality-form pair, take feasible x, p . Recall $s = Ax - b \geq 0$ and $r = c - A^\top p \geq 0$.

$$\begin{aligned}
 c^\top x - b^\top p &= c^\top x - p^\top Ax + p^\top Ax - p^\top b \\
 &= (c - A^\top p)^\top x + p^\top (Ax - b) \\
 &= \underbrace{\sum_j r_j x_j}_{\text{primal variable} \times \text{dual slack}} + \underbrace{\sum_i p_i s_i}_{\text{dual variable} \times \text{primal slack}}.
 \end{aligned}$$

Optimal pair $\Rightarrow$ zero products: strong duality makes the gap zero. Every summand is nonnegative, so every product must vanish.

Zero products $\Rightarrow$ optimal pair: the gap is zero. The two feasible objective values agree, so weak duality proves optimality.

Complementary slackness: the general theorem

Primal: minimize $c^\top x$ subject to $Ax \geq b, x \geq 0$.

Dual: maximize $b^\top p$ subject to $A^\top p \leq c, p \geq 0$.

Theorem 4.5 — assumptions: x and p are feasible for their respective problems.

Conclusion: they are both optimal if and only if

$$\begin{aligned} p_i(a_i^\top x - b_i) &= 0 && \text{for every row } i, \\ x_j(c_j - A_j^\top p) &= 0 && \text{for every variable } j. \end{aligned}$$

These are the two pairings just derived: dual variable $\times$ primal slack, and primal variable $\times$ dual slack.

Read each zero product carefully

For the inequality-form pair on the preceding page, take an **optimal feasible pair**. An inequality is **active** when its slack is zero.

Condition	Consequence
$p_i > 0$	Primal constraint i is active: $s_i = 0$.
$s_i > 0$	The corresponding dual variable is zero: $p_i = 0$.
$x_j > 0$	Dual constraint j is active: $r_j = 0$.
$r_j > 0$	The corresponding primal variable is zero: $x_j = 0$.

Zero slack does not force the variable to be zero. If $s_i = 0$, p_i may be zero or positive; if $r_j = 0$, x_j may be zero or positive.
Both factors may be zero. A zero variable does not force positive slack either.

Check complementary slackness in the example

Primal: minimize $13x_1 + 10x_2 + 6x_3$ subject to

$$5x_1 + x_2 + 3x_3 = 8, 3x_1 + x_2 = 3, x_1, x_2, x_3 \geq 0.$$

Dual: maximize $8p_1 + 3p_2$ subject to

$$5p_1 + 3p_2 \leq 13, p_1 + p_2 \leq 10, 3p_1 \leq 6; p_1, p_2 \text{ unrestricted.}$$

The feasible solutions $x^* = (1, 0, 1)$, $p^* = (2, 1)$ give:

Primal variable	x_j^*	Dual slack r_j	Product $x_j^* r_j$
x_1	1	$13 - [5(2) + 3(1)] = 0$	0
x_2	0	$10 - (2 + 1) = 7$	0
x_3	1	$6 - 3(2) = 0$	0

All three products are zero, so these feasible solutions are optimal. Dual constraints 1 and 3 are active.

The primal constraints are equalities: their slacks are already zero, and the dual variables are unrestricted.

Recover the dual variables from the positive variables

Suppose we know the optimal primal point $x^* = (1, 0, 1)$. Complementary slackness gives

$$\begin{aligned}x_1^* > 0 &\Rightarrow 5p_1 + 3p_2 = 13, \\x_3^* > 0 &\Rightarrow 3p_1 = 6.\end{aligned}$$

$$p_1 = 2, \quad p_2 = 1.$$

Finally check the remaining dual inequality: $p_1 + p_2 = 3 \leq 10$. Solving these equalities proposes a candidate; this last feasibility check completes the certificate.

The matrix equation packages the same two equalities

The basic variables are x_1, x_3 , so their columns and costs are

$$B = \begin{pmatrix} 5 & 3 \\ 3 & 0 \end{pmatrix}, \quad c_{\mathcal{B}} = \begin{pmatrix} 13 \\ 6 \end{pmatrix}.$$

$$B^{\top} p = c_{\mathcal{B}} \iff \begin{cases} 5p_1 + 3p_2 = 13, \\ 3p_1 = 6. \end{cases}$$

This is the equation we just solved. Equivalently, $p^{\top} = c_{\mathcal{B}}^{\top} B^{-1}$. No explicit inverse is needed here.

When all basic variables are positive, their active dual constraints determine the dual variables uniquely.

The simplex sign test checks the proposed dual solution

The appendix proof uses the same equation $B^\top p = c_B$. For any chosen basis, it gives a candidate dual solution.

$$\bar{c}_j = c_j - p^\top A_j.$$

Check every reduced cost: all must be nonnegative for this candidate to be dual feasible.

A basis determines a candidate, not necessarily a certificate. Even at an optimal degenerate point, a different basis may fail this sign check.

The next example shows why matching values alone is insufficient.

Optional: one basis gives paired primal and dual dictionaries.

A candidate dual solution must satisfy every constraint

Primal: minimize $13x_1 + 10x_2 + 6x_3$ subject to

$$5x_1 + x_2 + 3x_3 = 8, \quad 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.$$

Dual: maximize $8p_1 + 3p_2$ subject to

$$5p_1 + 3p_2 \leq 13, \quad p_1 + p_2 \leq 10, \quad 3p_1 \leq 6; \quad p_1, p_2 \text{ unrestricted.}$$

Another primal feasible basic solution is $x = (0, 3, 5/3)$, with cost 40.

$$5(0) + 3 + 3(5/3) = 8, \quad 3(0) + 3 = 3.$$

If this solution were optimal, $x_2, x_3 > 0$ would require dual constraints 2 and 3 to be active.

$$p_1 + p_2 = 10, \quad 3p_1 = 6 \quad \Rightarrow \quad p = (2, 8).$$

This is only a candidate. Does it satisfy the first dual constraint too?

The missing dual inequality rejects that candidate

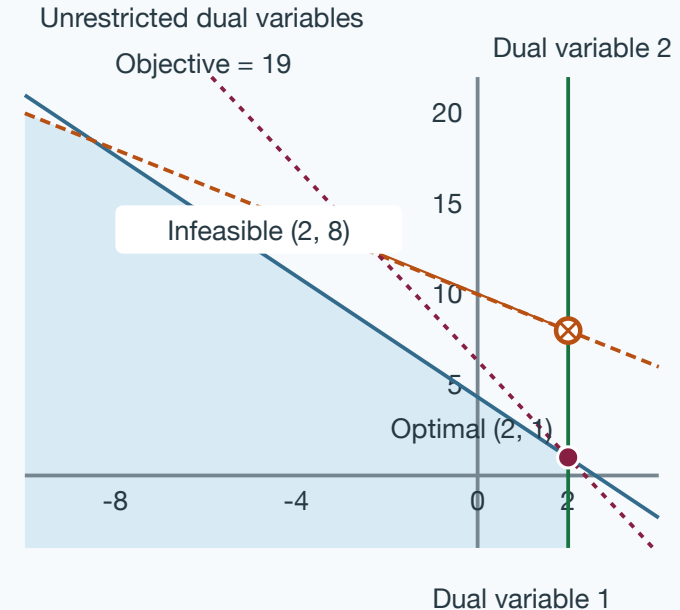
At $p = (2, 8)$, the first dual inequality fails:

$$5p_1 + 3p_2 = 34 > 13.$$

Its dual slack, and the reduced cost of x_1 , is $13 - 34 = -21$.

Although $b^\top p = 40 = c^\top x$, this is not a valid lower bound: p is infeasible.

Equal values and complementary products cannot replace the feasibility checks.



Solid blue: constraint 1; dashed orange: constraint 2; green vertical: constraint 3; dotted maroon: objective 19. The feasible region continues beyond the window.

Practice: construct and check a complete certificate

$$\begin{array}{ll} \min & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0. \end{array}$$

Try the primal point $x = (4, 0)$.

1. Write the one-variable dual.
2. Use the positive primal variable to find a candidate dual solution.
3. Check both feasibility and equality of the objective values.

Then check both kinds of complementary products.

Practice answer: one dual variable certifies a cost of 8

$$\max 4p \quad \text{s.t.} \quad p \leq 2, \quad p \leq 3, \quad p \geq 0.$$

Since $x_1 = 4 > 0$, its dual constraint is active: $p = 2$. This satisfies every dual constraint.

$$2(4) + 3(0) = 8 = 4(2).$$

Both points are feasible and their values match, so both are optimal.

$$p s = 2(4 - 4) = 0, \quad x_1 r_1 = 4(2 - 2) = 0, \quad x_2 r_2 = 0(3 - 2) = 0.$$

Use three checks before claiming optimality

1. Primal feasibility. Check every original constraint and variable sign.

2. Dual feasibility. Check every dual-variable sign and dual constraint.

3. Matching values. Compare objective values, or check all complementary products.

Bertsimas and Tsitsiklis §4.3, Theorems 4.4–4.5; Example 4.6.

Next: interpret these dual variables as prices and use weighted constraints to certify infeasibility.

Optional study: complete proofs, degeneracy, and additional examples

Optional proofs and extensions

Example 4.6: feasibility checks and dual plot

Strong duality: matrix proof and extensions

Paired primal and dual dictionaries

Example 4.5: both problems can be infeasible

Complementary slackness: necessity and sufficiency

Why degeneracy affects recovery of dual variables

An inactive constraint and its zero dual variable

Feasible-set size versus objective boundedness

Check the proposed primal solution explicitly

Problem C

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \quad 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

For the primal $\min 13x_1 + 10x_2 + 6x_3$, use $x^* = (1, 0, 1)$.

$$\begin{aligned}5(1) + 0 + 3(1) &= 8, \\ 3(1) + 0 &= 3, \\ x^* &\geq 0.\end{aligned}$$

$$c^\top x^* = 13(1) + 10(0) + 6(1) = 19.$$

The primal is feasible with cost 19. The next page checks the proposed dual solution.

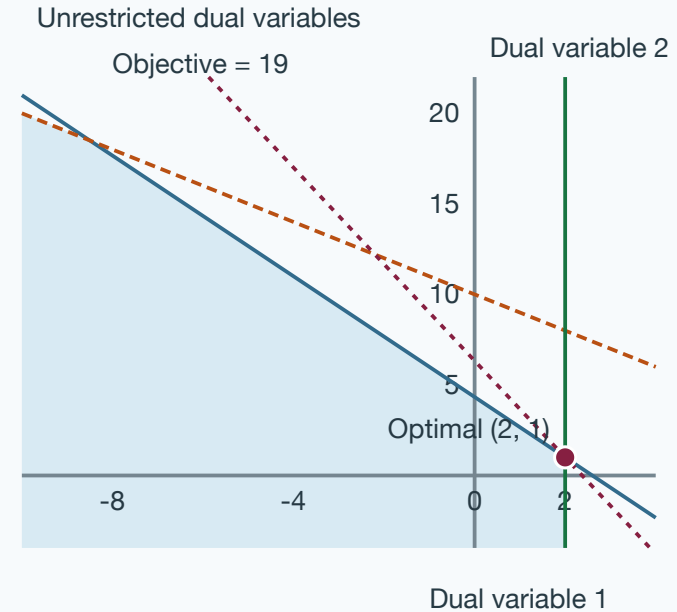
Check the proposed dual solution explicitly

Try $p^* = (2, 1)$. The dual maximizes $8p_1 + 3p_2$, with unrestricted dual variables.

$$\begin{aligned} 5(2) + 3(1) &= 13 \leq 13, \\ 2 + 1 &= 3 \leq 10, \\ 3(2) &= 6 \leq 6. \end{aligned}$$

$$b^\top p^* = 8(2) + 3(1) = 19.$$

The point lies in the dual feasible region. Its lower bound matches the primal cost.



Solid blue: constraint 1; dashed orange: constraint 2; green vertical: constraint 3; dotted maroon: objective 19. The feasible region continues beyond the window.

Strong duality proof: start with a terminating simplex run

$$\begin{aligned}(P) \quad & \min c^\top x && \text{s.t. } Ax = b, \ x \geq 0, \\(D) \quad & \max b^\top p && \text{s.t. } A^\top p \leq c, \ p \text{ unrestricted.}\end{aligned}$$

First assume A has full row rank and the primal is feasible with a finite optimum.

Phase I supplies a feasible basis. Run Phase II with a valid anticycling rule. The LP is not unbounded, so the run terminates at a feasible basis whose reduced costs are nonnegative.

$$x_{\mathcal{B}} = B^{-1}b \geq 0, \quad x_{\mathcal{N}} = 0, \quad c^\top - c_{\mathcal{B}}^\top B^{-1}A \geq 0.$$

Basic values may be zero. The argument uses the final reduced-cost signs, not a nondegeneracy assumption.

Strong duality proof: the final basis gives a feasible dual solution

Define p by $B^\top p = c_{\mathcal{B}}$, or equivalently $p^\top = c_{\mathcal{B}}^\top B^{-1}$.

$$c^\top - p^\top A = c^\top - c_{\mathcal{B}}^\top B^{-1} A \geq 0.$$

$$A^\top p \leq c.$$

These are exactly the dual constraints. Equality rows impose no sign restriction on p .

Every basic dual inequality is active. The nonbasic inequalities hold because the final reduced costs are nonnegative.

Strong duality proof: the objective values agree

At this feasible basic solution, $x_{\mathcal{N}} = 0$ and $x_{\mathcal{B}} = B^{-1}b$. Therefore

$$\begin{aligned} b^{\top} p &= c_{\mathcal{B}}^{\top} B^{-1} b \\ &= c_{\mathcal{B}}^{\top} x_{\mathcal{B}} \\ &= c^{\top} x. \end{aligned}$$

We have primal feasibility, dual feasibility, and matching values. Weak duality proves that both solutions are optimal.

This proves strong duality for full-row-rank standard form, including degenerate optima.

Extend the proof: row signs and dependent equalities

Replace $a_i^\top x \geq b_i$ by $a_i^\top x - s_i = b_i$, $s_i \geq 0$. Its zero-cost slack column gives $-p_i \leq 0$, hence $p_i \geq 0$.

Replace $a_i^\top x \leq b_i$ by $a_i^\top x + s_i = b_i$. Its slack column gives $p_i \leq 0$. Equality rows keep p_i unrestricted.

For a feasible equality system, discard dependent rows until the remaining rows are independent. Set the dual variables for the discarded rows to zero; retain the other dual-variable values.

Both $A^\top p$ and $b^\top p$ are unchanged by that extension, so the same certificate works for the original system.

Extend the proof: variable signs and the objective convention

For a free variable, write $x_j = u_j - v_j$, with $u_j, v_j \geq 0$. The two dual inequalities are

$$A_j^\top p \leq c_j, \quad -A_j^\top p \leq -c_j,$$

which together say $A_j^\top p = c_j$.

For $x_j \leq 0$, substitute $x_j = -u_j$. The dual inequality becomes $A_j^\top p \geq c_j$.

These substitutions preserve feasible solutions and objective values. They recover the general dual sign rules from standard form.

For a maximization primal, negate the objective to use the minimization proof, then undo that sign change. Dualizing the dual recovers the original problem, giving the symmetric theorem.

One basis gives paired primal and dual dictionaries

For $\min\{c^\top x : Ax = b, x \geq 0\}$, let $B = A_{\mathcal{B}}$, with remaining columns $A_{\mathcal{N}}$. Define

$$\beta = B^{-1}b, \quad H = B^{-1}A_{\mathcal{N}}, \quad \rho = c_{\mathcal{N}} - H^\top c_{\mathcal{B}}, \quad v = c_{\mathcal{B}}^\top \beta.$$

Primal dictionary

$$x_{\mathcal{B}} = \beta - Hx_{\mathcal{N}}, \quad f = v + \rho^\top x_{\mathcal{N}}.$$

Use dual slacks $r = c - A^\top p \geq 0$. Substitute $B^\top p = c_{\mathcal{B}} - r_{\mathcal{B}}$ into the remaining constraints:

Dual dictionary

$$r_{\mathcal{N}} = \rho + H^\top r_{\mathcal{B}}, \quad g = b^\top p = v - \beta^\top r_{\mathcal{B}}.$$

For an optimal primal dictionary, $\beta, \rho \geq 0$. At $r_{\mathcal{B}} = 0$, $r_{\mathcal{N}} = \rho \geq 0$ and $g = v$. Every feasible $r_{\mathcal{B}} \geq 0$ gives $g = v - \beta^\top r_{\mathcal{B}} \leq v$, so this dual dictionary is optimal.

A primal pivot exchanges the corresponding dual slacks

Suppose x_j enters and x_i leaves. Write $h = H_{ij} \neq 0$.

Primal row: solve for x_j instead of x_i .

$$x_i = \beta_i - hx_j - \sum_{k \in \mathcal{N} \setminus \{j\}} H_{ik} x_k.$$

Paired dual row: solve for r_i instead of r_j .

$$r_j = \rho_j + hr_i + \sum_{k \in \mathcal{B} \setminus \{i\}} H_{kj} r_k.$$

Thus r_i enters and r_j leaves the dual slack basis. Indices here follow the original variable labels.

Feasibility matters: intermediate dual dictionaries may be infeasible. At primal termination, nonnegative reduced costs make the paired dual solution feasible.

Example 4.5: both the primal and dual can be infeasible

In this example, the primal variables and the dual variables are unrestricted.

Primal: minimize $x_1 + 2x_2$ subject to $x_1 + x_2 = 1$, $2x_1 + 2x_2 = 3$.

Dual: maximize $p_1 + 3p_2$ subject to $p_1 + 2p_2 = 1$, $p_1 + 2p_2 = 2$.

The primal equations disagree: twice the first left-hand side would have to equal both 2 and 3.

The dual equations also disagree: their identical left-hand sides would have to equal both 1 and 2.

Thus primal infeasibility alone does not imply dual unboundedness.

Complementary slackness proof: both directions

For feasible x, p in the inequality form, set $s = Ax - b \geq 0$, $r = c - A^\top p \geq 0$.

$$c^\top x - b^\top p = \sum_j x_j r_j + \sum_i p_i s_i.$$

Necessity. If both points are optimal, strong duality says the gap is zero. Every summand is nonnegative, so each product must be zero.

Sufficiency. If every product is zero, the gap is zero. The two feasible objective values agree; weak duality makes both points optimal.

For equality rows the primal slack is identically zero; their unrestricted dual variables cause no extra condition. The variable–dual–slack terms give the same conclusion.

A zero basic variable need not make its dual constraint active

$$\min x_1 \quad \text{s.t. } x_1 = 0, x_1 \geq 0.$$

The only basic solution is $x_1 = 0$: it is optimal and degenerate. Its dual is

$$\max 0p \quad \text{s.t. } p \leq 1, \quad p \text{ unrestricted.}$$

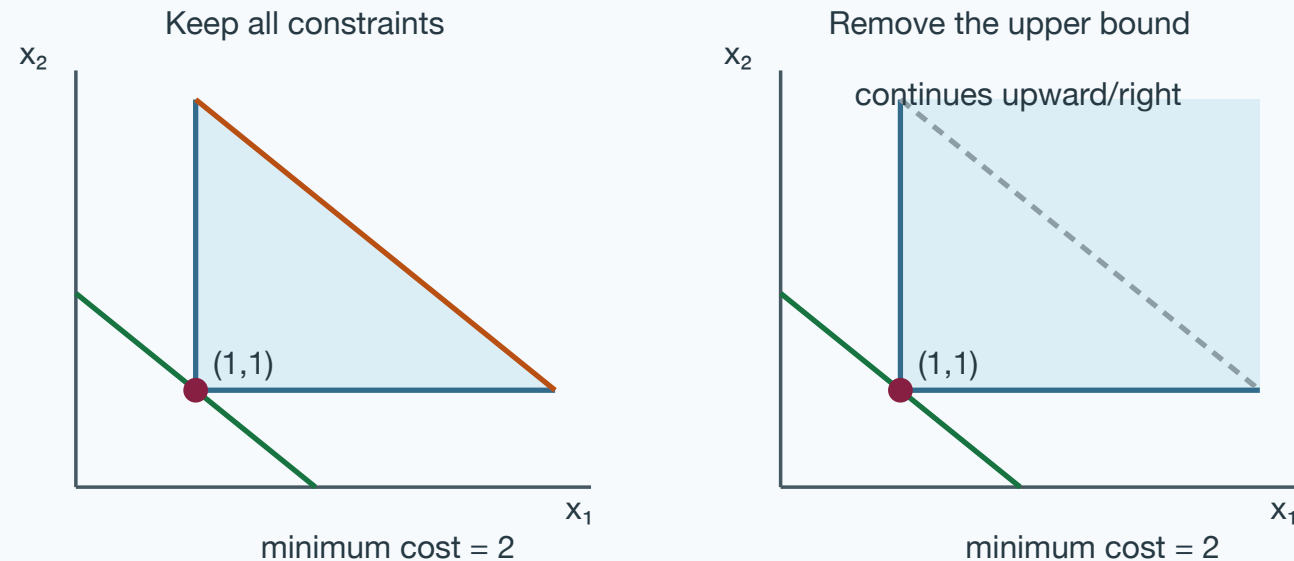
Complementary slackness says $0(1 - p) = 0$, which holds for every dual feasible p . It does not determine a unique dual solution.

The basis equation $B^\top p = c_B$ chooses $p = 1$, one valid optimal certificate. Strong duality still holds.

In a nondegenerate optimum, every basic variable is positive, so its dual constraint must be active; that is why the recovery argument is stronger there.

An inactive constraint has a zero dual variable at optimum

$$\min x_1 + x_2 \quad \text{s.t. } x_1 \geq 1, x_2 \geq 1, x_1 + x_2 \leq 5.$$



At $(1, 1)$, the upper bound has slack 3 and its corresponding optimal dual variable is zero. Removing that bound enlarges the region but leaves the minimum 2.

Why removing one inactive inequality preserves this optimum

Let x^* be optimal and suppose one inequality is strict at x^* . Remove that inequality.

If a newly allowed point y had lower cost, then $x(t) = (1 - t)x^* + ty$ would also have lower cost for every $t > 0$.

For sufficiently small $t > 0$, the removed inequality would still hold because it had positive slack at x^* . All other constraints hold along the segment by convexity.

That would give a cheaper point in the original feasible set, contradicting optimality.

This explains the zero dual variable: the inactive inequality is unnecessary for a matching lower-bound certificate.

An unbounded feasible set is not an unbounded objective

Clark's theorem: unless both members of a primal–dual LP pair are infeasible, at least one has an unbounded feasible set. See textbook Exercise 4.21.

For example, $\min\{x : x \geq 0\}$ has an unbounded feasible ray but a finite optimum 0.

“Unbounded feasible set” concerns where feasible points can go. “Unbounded minimization problem” means feasible objective values decrease without limit. These statements are different.

This result is additional theory; the optimality certificate still uses the same three checks.